

Theory Assignment 1

GNR 638: Machine Learning for Remote Sensing II

Anupam Rawat
IIT Bombay
22b3982@iitb.ac.in

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Problem Statement

Prove that the difference of two convex functions is not necessarily a convex function

Theory

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is called convex if for all $x_1, x_2 \in \mathbb{R}^n$ and for any $\lambda \in [0, 1]$, we have:

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2).$$

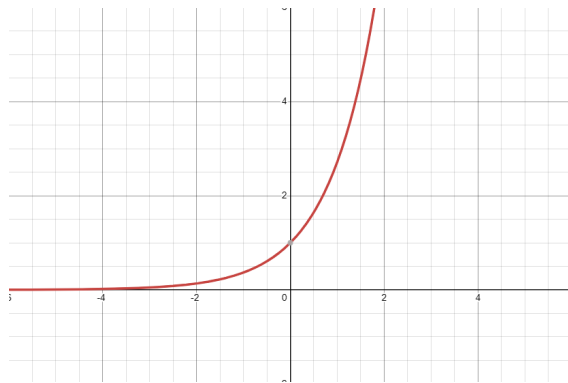
If the double derivative of $f(x)$ exists, then $f''(x) \geq 0, \quad \forall x \in \mathbb{R}^n$

Proof

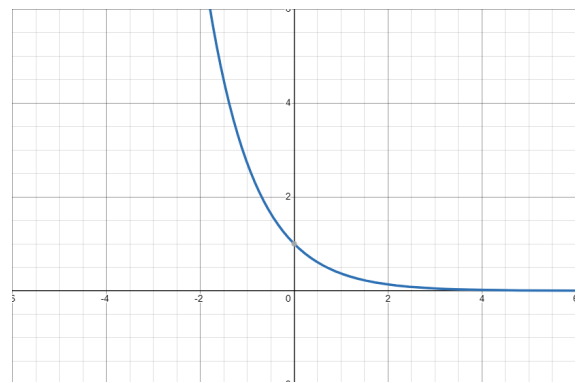
To disprove a statement, we can disprove by providing a counter-example. We can choose

$$f(x) = e^x$$

$$g(x) = e^{-x}$$



(a) Plot of $f(x)$



(b) Plot of $g(x)$

Figure 1: Comparison of $f(x)$ and $g(x)$

Both functions are differentiable at all points. The proof below shows that both $f(x)$ and $g(x)$ are convex:

$$f''(x) = e^x > 0$$

$$g''(x) = e^{-x} > 0$$

Let, $h(x)$ be defined as the difference of $f(x)$ and $g(x)$:

$$h(x) = f(x) - g(x) = e^x - e^{-x}$$

$$h'(x) = e^x + e^{-x} = \cosh(x)$$

$$h''(x) = e^x - e^{-x} = \sinh(x)$$

$$h''(x) = \begin{cases} \sinh(x) < 0, & \text{if } x < 0, \\ \sinh(x) > 0, & \text{if } x > 0, \\ \sinh(x) = 0, & \text{if } x = 0 \end{cases}$$

Since, double derivative of $h(x)$ is not always greater than 0, $h(x)$ is not convex, below is the graph for the same.

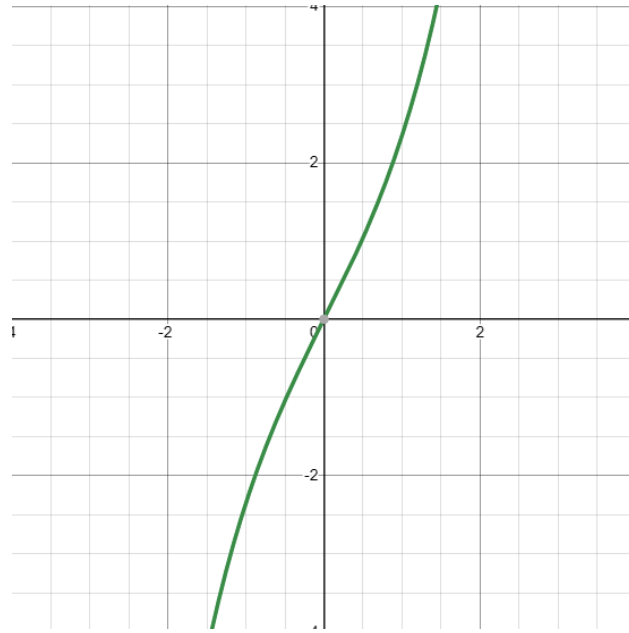


Figure 2: Plot of $h(x) = e^x - e^{-x}$

Hence Proved, that the difference of two convex functions is not necessarily convex